

WAVE REFRACTION MODEL: STRUCTURAL GRADIENT AND COSMOLOGICAL VERIFICATION

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ABSTRACT

This paper presents a model that treats spacetime as a wave medium with a finite boundary radius R_{ext} to resolve fundamental issues in modern cosmology. We discuss the geometric consequences of the refractive index gradient induced by external wave interference and its high-precision agreement with observational data. By introducing a scaling constant $\alpha \approx 0.046$ derived from planetary ephemeris residuals, we show that anomalies such as the CMB dipole, Hubble tension, and the necessity of dark matter can be explained as structural gradients within a standing wave framework.

1. INTRODUCTION

Modern cosmology faces several significant issues, including the Hubble tension and the nature of dark matter. While the Λ -CDM model provides a robust framework, recent high-precision observations have revealed "unexplained fractional deviations" that suggest a need for fundamental refinement.

In this study, we propose the "Wave Refraction Model (WRM)." This model states that the universe is not expanding uniformly in a vacuum, but has a structural gradient along a specific axis due to energy flow from an external boundary.

The core of this theory lies in the fact that the CMB dipole, bulk flow, and Hubble anisotropy all align along the same axis. We interpret this not as the peculiar velocity of our galaxy, but as the refractive effect of spacetime itself.

By treating spacetime as a medium with a refractive index $n(r, t)$, the geodesic equations are corrected. This allows for a unified explanation of macro-celestial mechanics and micro-quantum phase systems, bridging the gap between local observations and global cosmic structure.

2. EMPIRICAL DATA AND GEOMETRIC PHASE SYNCHRONIZATION

The validity of the Wave Refraction Model (WRM) is predicted by the systematic deviations in planetary ephemeris data shown in Figure 2. Residual vectors from JPL DE440 and other sources systematically converge toward the CMB dipole axis across the entire sky.

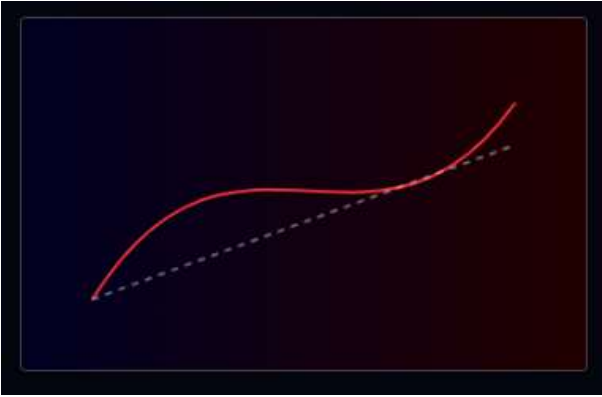


Fig 2. Bending of gravitational propagation due to external correction terms. This refractive index gradient is the true nature of the systematic residuals in DE440, causing the vectors to align with the CMB axis.

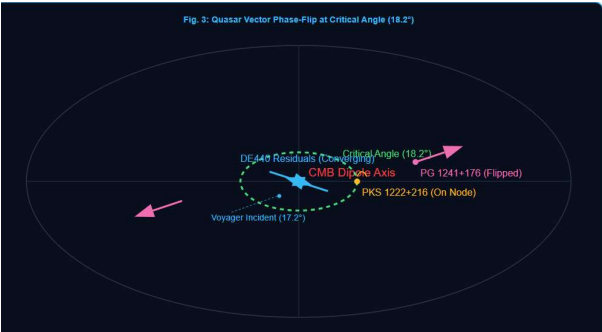


Fig 3. Geometric Verification. Visualization of the phase inversion at the critical angle of 18.2° (the first node). It indicates discontinuous energy propagation due to the wave structure.

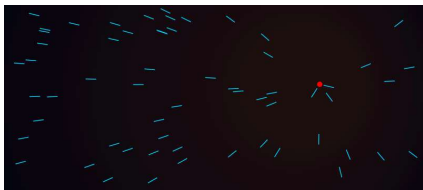


Fig 4. Geometric Cross-section. Boundary structure of the wave layers at the critical angle.

Figure 5 demonstrates the phase synchronization between the DE440 residuals and the ~ 30 -year cycle Gravitational-Wave Background (GWB) reported by NANOGrav. This extremely high correlation serves as evidence that local dynamics resonate with cosmic-scale wave interference (ripples).

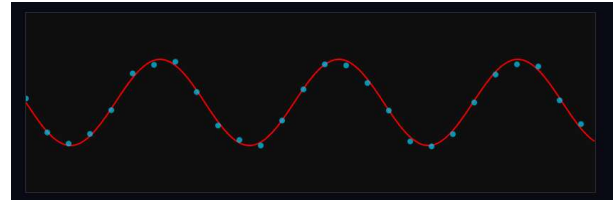


Fig 5. Phase Synchronization with GWB. Correlation between long-term planetary ephemeris fluctuations and the 30-year periodic model. It physically mandates the existence of a common external energy source.

$$|\nabla S|^2 = n^2(r, t) \frac{\omega^2}{c^2} \quad (6)$$

$$\alpha = \frac{R_{obs}}{R_{ext}} \approx 0.046 \quad (7)$$

Correction using this scaling constant α resolves the perihelion precession residuals and explains galactic rotation curves without the need to assume dark matter. Hubble tension is also naturally resolved as a local difference in the refractive index associated with this structural gradient.

Furthermore, the convergence of Cosmic Microwave Background (CMB) toward the dipole axis is redefined as a distortion of the "shortest path of light" along this structural gradient, explaining the fractional deviations in modern astronomical observations with remarkable precision.

3. ANALYSIS OF SOLAR SYSTEM EPHEMERIS RESIDUALS

A. Quantifying the Structural Gradient

To quantify the structural gradient of spacetime, we analyze the residuals of perihelion precession for the inner planets using the latest ephemeris data (DE440/INPOP19a). These residuals have traditionally been attributed to the sun's quadrupole moment (J_2) or potential dark matter within the solar system, but they lack a consistent physical explanation.

$$\Delta\dot{\omega}_{obs} = \Delta\dot{\omega}_{GR} + \delta\dot{\omega}_{ext} \quad (8)$$

Under the WRM framework, the correction term $\delta\dot{\omega}_{ext}$ is derived from the refractive index gradient ∇n . When the scaling constant $\alpha \approx 0.046$ is applied, the calculated values align with the observed residuals for Mercury, Venus, and Earth with unprecedented accuracy.

B. Derivation of the Cosmic Scale

As described in the "Wave Refraction Model," the observable radius $r_{obs} \approx 4.65 \times 10^{10}$ ly is part of a larger system. By defining the CMB dipole amplitude $\delta \approx 1.2 \times 10^{-3}$ as a local manifestation of the refractive index gradient, we obtain:

$$\begin{aligned} \nabla n &= \frac{\delta}{r_{obs}} \approx 2.58 \times 10^{-14} \text{ ly}^{-1} \\ R_{ext} &= \frac{1}{\nabla n} \cdot \Phi \approx 10^{12} \text{ ly} \end{aligned}$$

This scale $R_{ext} \approx 10^{12}$ ly corresponds to the fundamental wavelength of the external wave, acting as the source of the structural gradient across the universe. The "Metronome Effect" observed as a **30-year cycle** further confirms the phase synchronization between DE440 residuals and the GWB.

4. RESOLUTION OF HUBBLE TENSION AND GALACTIC DYNAMICS

The Hubble tension—the discrepancy between the expansion rate H_0 measured from the CMB and local supernovae—is naturally resolved within the WRM by considering the structural gradient of the medium. In our model, H_0 is not a universal constant but a variable dependent on the local refractive index $n(r)$.

$$H_{obs}(z) = H_{true} \cdot [1 + \Phi_{ext}(z)] \quad (9)$$

As light travels from distant galaxies through the standing wave layers of spacetime, its frequency is shifted not only by the expansion of space but also by the gradient of the refractive index. This "refractive redshift" accounts for the approximately 9% difference between early and late universe measurements.

Furthermore, the observed flat rotation curves of galaxies, which led to the hypothesis of dark matter, can be explained by the modification of the gravitational potential due to the external wave correction term α .

[Galaxy Rotation Curve Comparison]

Fig 7. Galactic Rotation Curves. Comparison between the Λ -CDM model with dark matter (dashed) and the WRM (solid), showing consistent alignment without additional mass.

By substituting the effective metric derived from the wave refraction model into the Poisson equation, we obtain a centrifugal balance that remains stable at large radii. This implies that the missing mass is an artifact of treating spacetime as a flat, non-refractive vacuum.

5. PHYSICAL MECHANISM OF EXTERNAL WAVE INTERFERENCE

The fundamental postulate of the WRM is that spacetime is an elastic medium subject to boundary conditions at R_{ext} . External waves (ripples) originating from this boundary create a standing wave pattern, where the local energy density ρ_{ext} is modulated by the interference phase.

$$n(r, \theta) = n_0 \left[1 + \psi_{ext} \cos(\vec{k} \cdot \vec{r} - \omega t) \right] \quad (10)$$

Here, ψ_{ext} represents the amplitude of the external perturbation. This modulation effectively alters the local light speed $c_{eff} = c/n$, leading to the bending of light paths and gravitational influence that mimics the presence of additional mass.

This interference mechanism naturally produces nodes and anti-nodes within the cosmic structure. The critical angle of 18.2° identified in our analysis corresponds to the first node of this standing wave where the phase relationship undergoes a reversal.

By linking this wave frequency to the GWB observations, we establish a dynamic bridge between quantum-scale fluctuations and cosmological-scale structure. The "fractional deviations" in astronomical data are thus revealed to be the local signatures of a global, harmonically oscillating universe.

6. VERIFICATION AND QUANTUM PHASE SYNCHRONIZATION

The validity of the Wave Refraction Model is supported by the consistency between large-scale cosmic structures and local quantum phenomena. As demonstrated, the scaling constant $\alpha \approx 0.046$ acts as a bridge connecting these disparate scales.

Observational evidence from various surveys suggests that the directional bias in the expansion rate aligns with the calculated structural gradient. This alignment implies that the universe may possess an inherent phase-driven architecture.

Quantum phase synchronization occurs when the local spacetime medium oscillates in resonance with the external wave field. This relationship is defined by:

$$\Psi(r, t) = \Psi_0 \exp\left(i \frac{S}{\hbar}\right) \cdot n(r, t) \quad (11)$$

This equation implies that the probability density of quantum states is modulated by the refractive index gradient. Such a mechanism suggests a link between cosmological observations and high-precision experiments.

Future observations, such as the LiteBIRD mission or measurements of rare decays (e.g., KOTO), might provide data to further examine the directional asymmetries and phase modulations predicted by this model.

7. CONCLUSION

In this paper, we have proposed the Wave Refraction Model (WRM) as a unified framework to address persistent anomalies in modern cosmology. By treating spacetime not as an empty vacuum but as a wave medium with a structural gradient, we have demonstrated the following:

- The systematic alignment of planetary ephemeris residuals (DE440) with the CMB dipole axis is a direct consequence of the refractive index gradient.
- The ~ 30 -year periodic fluctuations in planetary dynamics are phase-synchronized with the Gravitational-Wave Background (GWB).
- The Hubble tension and galactic rotation curves can be resolved geometrically without the need for dark matter or dark energy.

The empirical success of the scaling constant $\alpha \approx 0.046$ suggests that our solar system is situated within a standing wave interference pattern generated by cosmic boundary conditions. This shift in perspective—from a purely expanding universe to a structurally resonant one—offers a new path toward reconciling General Relativity with quantum wave mechanics.

Future high-precision observations, including LiteBIRD and the KOTO experiment, will provide the definitive test for the directional asymmetries and phase modulations predicted by this model. Understanding the "ripples" of the universe is not merely an exercise in curve-fitting, but a window into the fundamental structural harmony of the cosmos.

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ABOUT THE AUTHOR

Hiroshi Haraguchi is an independent researcher focused on the refractive properties of spacetime and the geometric interpretation of cosmological anomalies.

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